Injury Prediction and Player Safety in the NFL

Non-Technical Report

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Datasets Used:

Video Review (NFL Punt Analytics)

Injury Record (NFL 1st and Future)

Concussions in the NFL

Topic Description:

This project uses publicly available National Football League (NFL) data to improve the health and safety of NFL players. This project will develop insights into common injuries and determine what aspects of the game increase a player's risk for injury, with a particular focus on concussions.

Section 1: Executive Summary

This project is designed to focus on improving the health and safety of National Football League (NFL) players and to develop insights into what injuries are most prevalent and what aspects of the game may increase the risk of player injury or a specific injury type. The results will be used to make data-driven recommendations to the NFL on what strategies will be most effective in minimizing the negative impact the sport has on its players' wellbeing during and after their careers.

This analysis included three datasets. The first, 'Video_Review' is from the "NFL Punt Analytics" dataset. This dataset is a collection of information that consists of reviewable video evidence that outlines the events causing a concussion during punt players in the NFL 2016-2017 season. The second dataset, "InjuryRecord" accounts for 105 lower-limbs injuries that occurred over two seasons during the regular NFL season and provides information on the surface the game occurred on and the number of days the player missed due to injury. Lastly, the "Concussion" dataset, contains a list of concussion that occurred in the NFL from the year 2012 to 2014.

Using different machine learning models, analysis of the datasets determined that concussions have a clear relationship with the number of snaps into the game when injury occurs, the week of the season it occurs, and a player's position. Given these results, the NFL can implement rule changes like shorter games/seasons and offer additional protection to specific positions where the rate of injury is higher. Analysis also showed that tackling, being blocked, and 'friendly fire' were major sources of concussions on punt plays. New coaching techniques or personnel changes, such as moving to 10 players instead of 11 on punts, could reduce the rate of concussions. Overall, this project will empower the NFL to better protect their players and reduce long-term health impacts the game has on them.

Section 2: Introduction & Problem Definitions

The NFL is known as one of the most dangerous sports when it comes to player health and safety, while playing the sport and after the conclusion of one's career. According to one research study, it was discovered that American football players, particularly linemen, were found to be prone to early ageing "characterised by premature burden of chronic disease and reduced healthspan" (Grashow et al., 2023). The NFL has worked to address these issues by introducing rule changes, guardian caps to prevent concussions, redesigning equipment, and procedural changes like the introduction of the dynamic kickoff. These changes have produced success, as the NFL stated in the 2024 season they saw "concussions decreased 17% compared to last season" (National Football League [NFL], 2025). Although improvements have been made, two lab-based studies and one on-field study replicating the use of guardian caps showed there was no difference in players who wore them versus those who did not (O'Malley, 2024).

This project's objective is to quantify and predict the factors in NFL games that contribute to player injury risk, specifically the risk for concussions and lower body injuries. In particular, the project will identify plays, positions, actions, and environmental factors associated with injury risk and quantify these factors to understand the severity of their impact. To do this, machine learning models and statistical methods were used. If a model produced a high accuracy or correlation, it would be considered to indicate a strong relationship between the target feature and that specific variable. For example, if a model produces a high accuracy when evaluating the relationship between rate of concussion (target) and number of snaps into the game the injury occurred (specific variable), it could be reported there was a relationship between the two variables. The results of this analysis will allow for the NFL to make more efficient and effective decisions when investing into injury prevention techniques for their players.

Section 3: Data Overview

The first dataset, "Video Review", is a collection of information that was created based on reviewable video evidence that outlines the events that resulted in a concussion during punt players in the NFL 2016-2017 season. The target, 'Primary_Impact_Type', outlines if the concussion occurred from the impact of Helmet-to-Helmet, Helmet-to-Body, or Helmet-to-Ground. The "InjuryRecord" dataset accounts for lower-limbs injuries that occurred over two seasons during the regular NFL season and provides information on the surface the game occurred on and the number of days the player missed due to injury (or how severe it was). The last dataset, "Concussion", contains a list of concussion injuries that occurred in the NFL from the year 2012 to 2014. The data includes features such as 'Position', 'Week of Injury', 'Games Missed', 'Reported Injury Type', 'Average Playtime Before Injury'. All three datasets were accessed online through public downloadable files posted on Kaggle. The "Video_Review" and "InjuryRecord" datasets were posted by the NFL as a part of their Analytics Competition and the "Concussion" dataset was posted by a Kaggle user who accessed the data through the data.world website.

Data description

Table 1: Summary statistics, data types, and amount of data for all three datasets

Feature	Number of Entries	Mean	Median	Mode
Primary_Impact_Type_encoded	39	1.054	Helmet-to-body	'Helmet-to-body' and 'Helmet-to-helmet'
Friendly_Fire	39	0.514	No	No
Surface	105	0.543	Synthetic	Synthetic
BodyPart	105	1.724	Knee	Knee
Reported_Injury Type	325	0.185	Concussion	Concussion
Total Snaps	325	33.071	30.0	17.0

Data quality assessment

Table 2: Data quality assessment in all three datasets

Dataset	Missing Value Percentage	Handling Approach
'Video_Review'	0%	None needed
'InjuryRecord'	0%	None needed
'Concussion'	8.11%	Drop the rows with missing values

In the datasets 'Video_Review' and 'InjuryRecord', the player's information is unidentifiable. These datasets utilize ID numbers to characterize who the player is without providing their names, teams, or any other information. However, the 'Concussion' dataset lists the player's name, team, position, and the game the injury occurred in. All these factors make it easy to determine who the player was and offer little privacy. This project dropped all identifiable information except position which was useful for reaching the project's objective.

The data collected was from league or public datasets, so it was assumed player's consent was obtained.

Section 4: Data Cleaning, Preprocessing, Exploratory Data Analysis (EDA)

During analysis, it was determined that all three datasets had no duplicate rows and no duplicate columns. Missing or blank entries were dropped in all three datasets. The reason for that, as seen in Table 2, is the percentages of the features with missing values were very small or equal to 0%. All the data seemed to be consistent and accurate with the expectations of the datasets.

Although not very common, the only outliers found were in the 'Video_Review' dataset. There was one instance where the data lists the primary impact type as "unclear" and there are only two instances of "helmet to ground" listed as the primary impact type. These are not the norm, as helmet-to-helmet or helmet-to-body are most frequent in the dataset.

During this analysis, histograms, line graphs, scatter plots, and correlation maps were created to understand the relationship, if any, between each dataset's target variable and the other variants. The histograms were insightful in learning the distribution of the data across each feature and helpful in identifying outliers or areas of focus for further analysis. For example, it was determined a player is more likely to get a concussion when they are tackling or being blocked. The impact typically is in the form of Helmet-to-Helmet or Helmet-to-Body. The results of helmet-to-helmet are not that surprising as the NFL has made a large effort to minimize plays that result in that type of hit, however it still is present in the game today. The line graphs and scatter plots were used to visualize trends in the datasets and provide another guide on where to focus further analysis on. Lastly, the correlation maps provided a summary of the relationship between all features in the dataset against themselves; a correlation value close to 1.0 indicates a perfect relationship, while a value closer to 0 indicates a poor or non-existent correlation.

Section 5: Modeling/Analysis

To complete this project, machine learning models were used. Machine learning models are designed to take in data and create algorithms that identify trends and make predictions on new unseen data. Numerous of these models were used throughout the project and the performance varied based on which dataset and type of model were used. The model types that were most insightful included Partial Least Squares Regression (PLSR)/Principal Component Regression (PCR) and Decision Trees/Random Forest regressions.

PLSR is a useful technique when there are many variables in a dataset and the goal is to predict a dependent variable. For example, in the "Concussion" dataset, PLSR would utilize the numerous variables in the dataset such as "Total Snaps", "Position", "Games Missed", etc. to predict the likelihood of a concussion occurring based on those factors. PCR works very similarly to PLSR by also organizing the numerous variables into smaller components to make a prediction. However, PCR focuses on summarizing the data, without considering the target variable. In this case, PLSR would organize components in a way that is most beneficial to predict concussion risk, while PCR would use components that summarize data best then use those to predict concussion risk. The Decision Trees and Random Forest Regressions are models that take in the dataset and break it down into smaller and smaller subsets, similar in structure to a family tree which branches into more specific sections. Random forest regressions are more complex than decision trees since they use multiple decision trees to make predictions.

Machine learning models are prone to overfit. In other words, the models will learn these specific datasets and be unlikely to perform well when more or different data is introduced to the model. To combat this, techniques called standardization and cross validation were used to help prevent overfitting. Standardization helps all values in the dataset to be evaluated on the same scale; for example, the value of 'total snaps' in the "Concussion" dataset may range up to 100,

while the “Week of Injury” will be a maximum of 18. The model may be biased towards a large number so standardization ensures the values are evaluated equally. Cross validation is used to test how well the model performs on new unseen data; it is a way to see if overfitting is occurring before implementation. All the models were evaluated using accuracy or Root Mean Squared Error (RMSE). RMSE evaluates how far the model’s predications were from the actual values.

Section 6: Results

All three datasets performed their best with the PLSR model. After applying cross validation and standardization to the datasets, the “Video_Review” dataset produced a RMSE of 0.8241, the “InjuryRecord” dataset produced a RMSE of 0.4926, and the “Concussion” dataset produced a RMSE of 0.3498. All these values were the lowest produced cross validation RMSEs out of all techniques used on the three datasets; the average cross validation RMSE for this technique across all three datasets is 0.5555. Because RMSE evaluates how far the model’s predictions were from the actual values, the closer the RMSE value is to 0 (i.e. the smaller the value is), the better the model performed.

The PLSR model most likely performed well with these datasets, more specifically the “InjuryRecord” and “Concussion” datasets, due to the high multicollinearity present in the datasets. Multicollinearity is when two or more variables in the dataset are highly correlated which results in the model struggling to differentiate between the variables. For example, in the “InjuryRecord” dataset, there are two variables called ‘PlayerKey’ and ‘PlayKey’. The ‘PlayerKey’ variable is a unique five-digit number that identifies the NFL player who was injured. The ‘PlayKey’ variable is a unique number that identifies a player’s play within the game. So, a ‘PlayerKey’ value could be 39873 and a ‘PlayKey’ value could be 39873-4-32; although the numbers are not exact, the ‘PlayerKey’ value is utilized in both variables so the

model could easily confuse both variables. The PLSR model can combat multicollinearity through a technique called dimension reduction. Dimension reduction reduces the number of features or variables in a model while keeping the important information. In other words, it helps remove some of the highly correlated variables and limits model confusion without preventing the model from learning trends and patterns. The "Video_Review" dataset's best RMSE being 0.8241 indicates that the dataset did not perform very well throughout the project. This dataset only consists of 37 entries. With such a small dataset size, it can be assumed the model is overfitting and struggling to perform well on unseen and new data that is being introduced.

The PCR model performed slightly worse than PLSR, with an average cross validation RMSE of 0.5899 across all three datasets. The Video_Review" dataset produced a RMSE of 0.9214, the "InjuryRecord" dataset produced a RMSE of 0.4972, and the "Concussion" dataset produced a RMSE of 0.3513. All these RMSE scores were slightly higher than what was seen in the PLSR models; to reiterate, the smaller the RMSE value, the better the model performed. However, these datasets still performed well compared to models that were not PLSR. As mentioned in the Modeling/Analysis section (Section 5), PCR focuses on summarizing the data without considering the target variable, where PLSR organizes the data in a way that is most beneficial to predict the target variable. Because the PLSR model performed better than the PCR model in all three datasets, it can be assumed that the target of each dataset was strongly correlated with the data (Scikit-learn, n.d.). More specifically, this means that this model does the best job in accurately predicting when concussions occur in the "Concussion" dataset, what surface lower limb injuries occurred on in the "InjuryRecord" dataset, and the primary impact type leading to concussions on punt plays in the "Video_Review" dataset.

When it comes to the Decision Tree and Random Forest regressions, the models were evaluated using cross validation accuracy and standard deviation. Cross validation is the same as it was in the other models; a way to test how well the model performs on new unseen data or its ability to generalize. Accuracy is how many correct predictions the model classifies divided by the total number of predictions made; the higher this number the better the model performed. Standard deviation is a measure of how far the data points are from the mean of the dataset. The lower this number is, the more consistent the data is. When using these models, the goal is to have the model learn what features or variables split the data most effectively; it helps find the features that help the model most accurately in predicting the target value (concussion, surface, or primary impact type). This concept is called feature importance.

The ‘Concussion’ dataset performed well with decision trees and random forest regressions; the decision tree produced at 90.62% cross validation accuracy with a standard deviation of 4.9% and the random forest regression produced an 84% cross validation accuracy and 4.7% standard deviation. The most important features in predicting concussions in these datasets were “Team”, “Season”, and “Average Playtime Before Injury” which had an importance level of about 0.27, 0.24, and 0.13 respectively. The higher the importance number, the more predictive the feature is. The next four features were “Games Missed”, “Total Snaps”, “Week of Injury”, and "Opposing Team” which all had an importance value of about 0.07. This information indicates that team identity and the NFL season were predictive patterns in this dataset. This could suggest that there are differences in team’s training procedures, coaching schemes and techniques or that rule changes and equipment redesign could be influencing the data. In addition, a player’s playtime also has a strong influence on their risk for concussion. It is not surprising that the longer a player is on the field, that their likelihood of a concussion

increases as there are more opportunities for the individual to be injured. However, comprehending when a player's playtime puts them into a high-risk situation, could allow the NFL to limit play time or consider shortening games. The 'Concussion' dataset is the largest of the three datasets with 390 entries. With a large dataset, these models are more likely to produce well balanced classes that are related to the target value (i.e. "Reported Injury Type") and therefore create a stable and strong model for this dataset.

The other two datasets did not perform as well. For the "Video_Review" dataset, the decision tree regression produced a 42.93% cross validation accuracy with a standard deviation of 16.7% and a cross-validation accuracy of 45.47% for random forest regression with a standard deviation of 17%. The most important features were "PlayID" and "Primary_Partner_GSISID" with both having an importance value of about 0.23. The "PlayID" feature is a numeric ID number that identifies the play the concussion occurred on; it would make sense that this feature would be highly influential in predicting the primary impact type that led to a concussion on punt plays. The "Primary_Partner_GSISID" is a unique player ID number that represents the impacting player involved in the helmet impact. Again, this directly correlates with the target variable of primary impact type so its importance value being higher is not surprising.

For the 'InjuryRecord', the decision tree produced a 42.40% cross validation accuracy with a 8.94% standard deviation and a 40.26% cross validation accuracy with a standard deviation of 9.18% for random forest regression. For this dataset, the top features were "PlayerKey" and "GameID" which both had an importance value of about 0.33. As explained above, 'PlayerKey' is a variable that is a unique five-digit number that identifies the NFL player who was injured. The "GameID" variable is a unique identifier of the game the player was playing in when injury occurred. Both features add information to the model about the game and

player who suffered a lower limb injury so it is understandable they would be influential in predicting outcomes.

Overall, these results indicate that models were not very accurate for either the "Video_Review" or "InjuryRecord" datasets and the data points showed a lot of variability, especially in the "Video_Review" dataset which produces large standard deviations. A major issue for both datasets is their size. The "Video_Review" dataset has 37 entries and the "InjuryRecord" dataset contains 105 entries. With such a small sample size, the models are most likely overfitting due to lack of features and entries, so it is not surprising they perform poorly. A poor model performance reiterates that the connection between the features and target value are not very strong in either dataset. Tables 3 and 4 provide a summary of the appropriate evaluation metrics for all machine learning models used in this project.

Table 3: *Summary Table of CV RMSE for various machine learning models*

Dataset	Polynomial Terms	Interaction Terms	Lasso Regression	Ridge Regression	Elastic Net Regression	Forward Selection	Backward Selection	PCR	PLSR
Video_Review	2.595	2.636	1.07	1.25	1.07	1.2076	1.2076	0.9214	0.8241
Injury_Record	1.256	0.712	0.51	0.54	0.51	0.5076	0.5208	0.4972	0.4926
Concussion	0.384	0.403	0.40	0.37	0.36	0.3721	0.3708	0.3513	0.3498

Table 4*: Summary Table of CV accuracy for various machine learning models*

Dataset	Logistic Regression	Support vector machine (with best C)	Decision tree regression	Random Forest Regression	KNN	Gradient Boosting
Video_Review	0.5946	0.7586	0.4293	0.4547	0.6250	0.5533
Injury_Record	0.5048	0.5357	0.4240	0.4026	0.4770	0.5346
Concussion	0.8031	0.9500	0.9062	0.8400	0.8460	0.8885

Section 7: Recommendations

The "Video_Review" and "InjuryRecord" datasets never really produced any strong results. The biggest problem with the datasets is their size. If more data was collected over a longer time, the conclusions could be more concrete and the recommendations for interventions would be more specific. However, based on the insights produced from the "Video_Review" dataset (the accuracy for this dataset was around 50-60% and had an average RMSE of 1.41), the indication is that features like friendly fire, player activity, and turnover related injuries have limited predictive value regarding primary impact type. By being unable to predict what factors

affect primary impact type, it is harder to recommend rule changes or promote different coaching techniques in preventing injury.

Similarly, after each analysis, it was often indicated that the features in the "InjuryRecord" dataset did not have a strong enough relationship to the target, so the model performed poorly (the highest accuracy was 65% across all models and lowest RMSE was 0.4926). This indicated that features like body parts injured, and length of time missed due to injury did not have a strong relationship with the surface the injury occurred on. As discussions regarding the dangers of synthetic turf continue in the NFL, this analysis disapproved the need to invest in natural grass and can save the league time and money on changing the field if there is no real prevention measure being achieved. Again, more data on the impact and severity of player's injuries needs to be collected for a definite recommendation to be made.

The "Concussion" dataset performed well in decision tree and regression models. This indicates that the target 'Reported Injury Type' has a clear relationship with features like number of snaps into the game when concussion occurs, a player's position, and the average playtime before injury. Based on these insights, implementing protective equipment for head injuries, specifically for cornerbacks, wide receivers, and safeties who experienced the most injuries in the 2012-2014 NFL seasons is vital. This could focus on re-designing helmets, exploring different tackling techniques, or even considering dissolving contact in the game of football and moving towards similar rules to flag football. Another option would be to shorten the NFL season or shorten the length of games to minimize the number of repetitive contact players face. Additionally, the NFL could use the data regarding average playtime prior to injury (i.e. a concussion) to comprehend when a player's playtime starts to put them at a high risk for injury. This could influence the NFL to put players on a playtime limit as it is not surprising that the

longer a player is on the field, that their likelihood of a concussion increases as there are more opportunities for the individual to be injured.

As stated throughout this report, all three datasets were small. The “Video_Review” dataset had 37 entries, the “InjuryRecord” dataset had 105 entries, and the “Concussion” dataset was the biggest with 390 entries. Small datasets mean the machine learning models are more prone to overfitting and produce less accurate results. In addition, the most recent dataset used was published in 2019. Having a full dataset from more recent seasons, especially since the adaption of the dynamic kickoff, helmet re-design, and new penalties for contact to players head and lower bodies, would provide the project with a more accurate insight and in turn recommendations.

Moving forward, implementing risk studies, A/B testing of coaching interventions, and longitudinal study of injury trends in the NFL would be beneficial. Risk studies would help determine and measure what actions are the riskiest in NFL games. A/B testing of coaching interventions would allow a direct comparison between two different approaches to see which is best in preventing injuries. Longitudinal studies would provide more information and accurate insight into how the NFL’s injury rates have changed over multiple years and identify areas of concern. An opportunity to also collect and analyze data regarding the NFL’s return-to-play policy, especially after a concussion, would bring more insight into the health and safety of players in the league. Understanding how the NFL clears their players to return to play and how these decisions affect the likelihood of repeated concussions, or the development of Chronic Traumatic Encephalopathy (CTE) would not only improve players’ wellbeing but could prevent the NFL from facing additional lawsuits from current and former players.

References

Abdi, H. (2010). Partial least squares regression and projection on latent structure regression (PLS Regression). *WIREs Computational Statistics, 2*(1), 97-106. **https://doi.org/10.1002/wics.51**

Grashow R., Shaffer-Pancyzk TV., Dairi I., Lee, H., Marengi, D., Baker, J., Weisskop, M. G., Speizer, F. E., Whittington, A. J., Taylor, H. A., Keating, D., Tenforde, D., Guseh, J. S., Wasfy, M. M., Zafonte, R., Baggish, A. (2023). Healthspan and chronic disease burden among young adult and middle-aged male former American-style professional football players. *British Journal of Sports Medicine, 57*(3)**,** 166-171.

Kaggle. (2021). Concussions in the NFL (2012-2014). *Kaggle.* **https://www.kaggle.com/datasets/rishidamarla/concussions-in-the-nfl-20122014/data**

Kaggle. (2019). NFL 1st and Future. *Kaggle*. **https://www.kaggle.com/competitions/nfl-playing-surface-analytics/data**

Kaggle. (2019). NFL punt analytics competition dataset. *Kaggle*. **https://www.kaggle.com/competitions/NFL-Punt-Analytics-Competition/overview**

Logistic Regression and regularization: Avoiding overfitting and improving generalization. (2023, January 5). Medium. Retrieved June 29, 2025. **https://medium.com/@rithpansanga/logistic-regression-and-regularization-avoiding-overfitting-and-improving-generalization-e9afdcddd09d**

NFL Football Operations. (2025). *Concussions Decrease to Historic Low in 2024 NFL Season*. National Football League. **https://operations.nfl.com/updates/football-ops/concussions-decrease-to-historic-low-in-2024-nfl-season/**

O'Malley, K (2025, February 7). *There's a way to deal with brain injuries in football. It isn't*

safety gear. Harvard Public Health. **https://harvardpublichealth.org/policy-practice/the-nfls-concussion-solutions-are-an-illusion/**

Scikit-learn. (n.d.). *Principal Component Regression vs Partial Least Squares Regression.* **https://scikit-learn.org/stable/auto_examples/cross_decomposition/plot_pcr_vs_pls.html**

The Pennsylvania State University (n.d.). *Combining Clusters in the Agglomerative Approach* [Lecture Notes]. **https://online.stat.psu.edu/stat505/lesson/14/14.4**

Appendix
Jupyter Notebook Links

Jupyter Notebook Topic	**Link to Notebook**
Preprocessing	https://github.com/hnm15/ModB/blob/cd4113c89dad2d7afab037c0bf91dddd932673b7/Week2.ipynb
Univariate Analysis	https://github.com/hnm15/ModB/blob/cd4113c89dad2d7afab037c0bf91dddd932673b7/Week4.ipynb
Bivariate Analysis	https://github.com/hnm15/ModB/blob/cd4113c89dad2d7afab037c0bf91dddd932673b7/Week6.ipynb
Multivariate Analysis	https://github.com/hnm15/ModB/blob/cd4113c89dad2d7afab037c0bf91dddd932673b7/Week9.ipynb
Basic Modeling	https://github.com/hnm15/ModB/blob/cd4113c89dad2d7afab037c0bf91dddd932673b7/Week11.ipynb
Linear Regression Part 1	https://github.com/hnm15/capstone/blob/8d6cc89716a283cc60a4ed782a07c702c3cf354f/Wk1-JupyterNotebook-Metcalf.ipynb
Linear Regression Part 2	https://github.com/hnm15/capstone/blob/fe79c2d4a529a3097d9db10da310041c0c4f90bf/Wk2-JupyterNotebook-Metcalf.docx.ipynb
Linear Regression Part 3	https://github.com/hnm15/capstone/blob/20d2b93736b1bb7b37b766d1b7e4cd1203f5a9f9/Wk3-JupyterNotebook-Metcalf.ipynb
Logistic Regression and Feature Scaling	https://github.com/hnm15/capstone/blob/2a5f2cc1ec6d00b7d950f4f1fcbdac93be7a1bf4/Wk4-JupyterNotebook-Metcalf.ipynb
Support Vector Machines	https://github.com/hnm15/capstone/blob/fe79c2d4a529a3097d9db10da310041c0c4f90bf/Wk5-JupyterNotebook-Metcalf.ipynb
Decision Trees and Random Forests	https://github.com/hnm15/capstone/blob/fe79

	c2d4a529a3097d9db10da310041c0c4f90bf/Wk6-JupyterNotebook-Metcalf.ipynb
K-Nearest Neighbors	https://github.com/hnm15/capstone/blob/9e1ca4752205cc1251a6a945807a6c406e6da906/Wk8-JupyterNotebook-Metcalf.ipynb
Gradient boost, learning rate, number of estimators, tree depth, and regularization	https://github.com/hnm15/capstone/blob/9e1ca4752205cc1251a6a945807a6c406e6da906/Wk9-JupyterNotebook-Metcalf.ipynb
K-means: Elbow method, Silhouette score, Feature scaling	https://github.com/hnm15/capstone/blob/9e1ca4752205cc1251a6a945807a6c406e6da906/Wk10-JupyterNotebook-Metcalf.ipynb
DBSCAN, HAC, Linkage methods in HAC, Dendrograms	https://github.com/hnm15/capstone/blob/9e1ca4752205cc1251a6a945807a6c406e6da906/Wk11-JupyterNotebook-Metcalf.ipynb